

Physical Inventory Counting

Front Royal, VA cold storage · Zebra MC9000-series handhelds

This is the working manual. It assumes you have never opened the app before and takes you from an empty deployment through to a finished count sent to the ERP. Read Part 1 and Part 2 once, when the app is first set up. Parts 3 to 5 are what you do for every count. Part 6 is the cycle-count programme, Part 7 is what to do when something goes wrong.

About the screenshots. Every picture below is the real application, photographed against a count part-way through a morning: the actual Front Royal bin list, three teams, four scanners, two aisles finished and three being counted. The names, pallet IDs and lot codes are made up. If a screen changes, `npm run docs` rebuilds every picture in this manual from scratch and re-prints [SOP.pdf](#) — the same manual typeset for paper.

Printed September 25, 2026

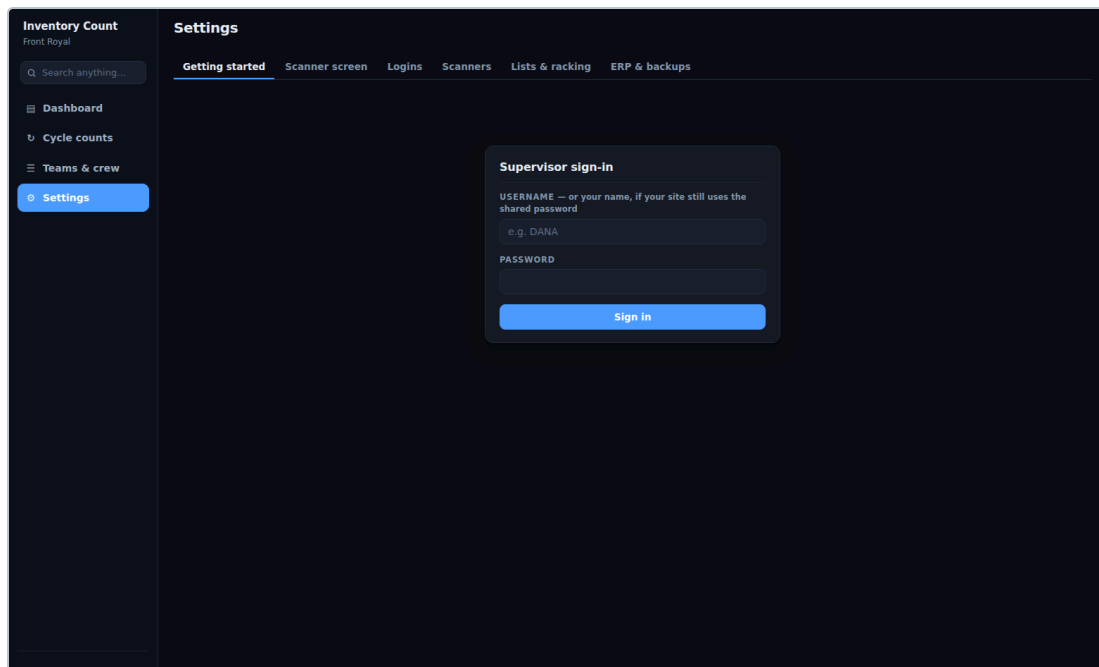
Contents

Part 0 — What the system is	3
Part 1 — First-time setup: getting in	5
Part 2 — First-time setup: the warehouse	8
Part 3 — Setting up a count	16
Part 4 — Counting day	21
Part 5 — Finishing a count	33
Part 6 — The cycle-count programme	39
Part 7 — When something goes wrong	41
Part 8 — Capacity: what the system will take	45
Appendix A — File column reference	46
Appendix B — Server settings reference	48

Part 0 — What the system is

One server, four screens. Everything below happens in a browser; nothing is installed on a PC.

SCREEN	ADDRESS	WHO USES IT	SIGN-IN
Scanner	/	Counters, on the handhelds	Each scanner has its own link
Dashboard	/admin	Supervisor running the count	Supervisor login
Settings	/settings	Whoever sets the count up	Supervisor login
Office board	/board	Anyone — put it on the office TV	None. Read-only
Cycle counts	/cycle	Whoever runs the daily programme	Supervisor login



Every supervisor page opens on the same sign-in. Signing in on one carries to all four.

What the system does. It holds a list of every bin in the warehouse and a list of what the ERP thinks is in them. Counters scan pallets into bins on the handhelds. The app compares the two, tells you where they disagree, sends people back to look again where it matters, and produces an adjustment file for the ERP.

What a "count session" is. One count — a wall-to-wall in October, a spot check of the freezer, the year's cycle-count programme. Everything (bin list, inventory report, team assignments, every scanned line) belongs to a session. You can have several open at once; scanners choose which one they are working on at sign-on.

Two words you need.

- **Aisle** — a rack run, e.g. `F01`. Teams are assigned whole aisles.
- **Block** — aisles that back onto each other share a **block**, and only one team may be active in a block at a time, so two crews never work opposite faces of the same racking. Pairing aisles into blocks is a one-time job (Part 2, step 7).

Bin codes. A code like `F01A001` reads as zone `F`, aisle `01`, level `A`, position `001`. Four positions per bay; odd positions are the front face, even are the back. The app derives the aisle from the code when your file has no aisle column.

Part 1 — First-time setup: getting in

Do this once, with whoever administers the server. Allow 20 minutes.

Step 1 — Deploy the server

On Railway (how this site runs). Deploy the repository; `railway.json` builds the Dockerfile, and the app listens on the port Railway gives it. Attach a **persistent volume** and point `DB_PATH` at it — see step 2 — or the database is wiped on every redeploy.

On a PC inside the warehouse (no internet needed).

```
ADMIN_PASSWORD='pick-something' docker compose up -d
```

Then give that PC a static IP or a DHCP reservation, because the handhelds are pointed at its address and that address must never change. Scanners reach it at `http://<that-ip>:3000/`.

Step 2 — Set the server settings

On Railway these are **Variables**; on a PC they go in `docker-compose.yml`. Set these before anybody signs in:

SETTING	SET IT TO	WHY
<code>ADMIN_PASSWORD</code>	A password only supervisors know	The shared way in, used once to create real logins
<code>DB_PATH</code>	A path on the persistent volume, e.g. <code>/data/inventory.db</code>	Otherwise a redeploy loses the count
<code>SITE_TIMEZONE</code>	<code>America/New_York</code>	Dates on reports and cycle-count due dates
<code>SUPERADMIN_USER</code>	e.g. <code>sitelead</code>	Creates one permanent admin login at start-up
<code>SUPERADMIN_NAME</code>	That person's name, as it should appear in the log	Shown against everything they do
<code>SUPERADMIN_PASSWORD</code>	A strong password, at least 8 characters	If unset it falls back to <code>ADMIN_PASSWORD</code>

The superadmin account is created when the server starts and cannot be deleted from inside the app, so you can never lock yourself out. The app refuses to create it if the password would be the default `changeme` or is shorter than 8 characters — it will say so in the start-up log and carry on without it.

Never put a password in the repository. These are server settings for a reason: anything committed to git is readable by anyone with access to the code, for ever.

Step 3 — Sign in for the first time

Open `/settings`. Sign in with the shared `ADMIN_PASSWORD` (leave the username box empty), or with the superadmin username and password if you set one.

Step 4 — Create a login for each supervisor

Settings → Logins → Supervisor logins.

1. Type the person's **name**, a **username** and choose their **role**:
 - **Admin** — can do everything, including managing logins.
 - **Supervisor** — runs counts; cannot add or remove logins.
2. Press **Add**. The app shows a **starter password** once — something like `winter-4k2p`. Write it down and hand it to them; it is not shown again.
3. The first time they sign in they are made to choose their own password before they can do anything. Nobody but them knows it after that.

If somebody forgets their password, an admin presses **Reset password** on their row and gives them the new starter password. The same forced change happens again.

Supervisor logins

shared password only

everyone who changes something is recorded by name in the log

The shared password is the only way in.

Add an admin login below, sign in as it to check it works, then set SHARED_PASSWORD_LOGIN=off.

Give each supervisor their own login. **Admins** can manage accounts and everything else; **supervisors** can do everything except manage accounts. Deactivating a login keeps its history in the log but stops it being used.

USERNAME	FULL NAME	PASSWORD	ROLE	
DANA	Dana Whitfield	leave blank to generate	Supervisor	Add login

Leave the password blank and one is generated for you — read it out once, and they choose their own the first time they sign in. Type one instead and they still have to replace it, so nobody but its owner ever knows a working password.

USERNAME	NAME	ROLE	STATUS	LAST SIGNED IN	ADDED
No logins yet — everyone is signing in with the shared password.					

Your password

You are signed in with the shared password, which has no account to change. Ask an admin for a login of your own.

Settings → Logins. Each person has their own login and their own password; the starter password is shown once.

Step 5 — Turn the shared password off

Once at least one admin login exists, set `SHARED_PASSWORD_LOGIN=off` in the server settings. From then on everybody signs in as themselves and the log names who did what.

The app will not let this lock you out: if there are no admin accounts, the shared password keeps working regardless of the setting.

Part 2 — First-time setup: the warehouse

Also once, though you will come back to steps 6 and 7 whenever the racking changes.

Step 6 — Upload the bin list

Settings → Lists & racking → Bin list.

This is every location in the warehouse. It is what a scanned bin is checked against, and it is what defines the aisles teams get assigned to. **Upload this before anything else.**

- For this site, press **Load the Front Royal bin list** — the ERP bin export ships with the app: racks F01-F24 and A01-A04 plus WIP, the NIL bin and other areas. Staging lanes and dock doors are counted manually and are deliberately left out.
- Otherwise choose a CSV or Excel file. Column names are matched loosely — see [Appendix A](#). The only column you must have is the bin location.
- **Replace what is there** wipes the existing list first. Leave it unticked to add to it.

The card reports what it read: how many bins, how many aisles, and anything it skipped.

Settings → Lists & racking. Each list has its own card, and each reports exactly what it read.

Step 7 — Pair the aisles into racking blocks

Settings → Lists & racking → Aisles & racking blocks.

Aisles that back onto each other must share a block so two teams are never working the same racking from both sides.

1. Set **Aisles per block** to 2 (most double-deep racking).
2. **Skip first** is 1 if the first aisle has a wall behind it rather than another aisle.

3. Press **Auto-pair aisles**, then read the table and correct anything that is wrong by typing a block name straight onto a row.

If you have uploaded a rack drawing, **Pair from drawing** uses it instead — more reliable than counting aisles off by hand.

Aisles & racking blocks

Aisles in the same **block** share racking, so only one team can be active in a block at a time. Pair them automatically, or type a block name on any row. Teams are queued onto these aisles on the [Dashboard](#).

AISLES PER BLOCK

2

SKIP FIRST

0

Auto-pair aisles

Pair from drawing

AISLE	BLOCK	BINS	COUNTED	PROGRESS	STATUS
Dry Dock - Aisle A01	A01+A02	132	0		—
Dry Dock - Aisle A02	A01+A02	258	0		—
Dry Dock - Aisle A03	A03+A04	258	0		—
Dry Dock - Aisle A04	A03+A04	36	0		—
Freezer - Aisle F01	F01+F02	654	652		done
Freezer - Aisle F02	F01+F02	618	240		active team 1 (ABCDEF)

Aisles paired into racking blocks. F01+F02 share racking, so only one team works that block at a time.

Step 8 — Register the scanners

Settings → Scanners → Scanner setup.

Each handheld gets its own link. Opening that link once is what signs the scanner in; after that the server accepts counts from it and stamps its name on every line.

- 1. Type a **scanner name** — use what is written on the device, e.g. `SCANNER-05`. Add notes (asset tag, "freezer unit") if it helps.
- 2. Press **Add scanner**. Repeat for every handheld.
- 3. Press **Print setup cards**. You get one card per scanner with its own QR code. Cut them up and tape one inside each cradle.

Scanner setup
scanners must sign in

each scanner gets its own link — open it once on the device and add it to the home screen

SCANNER NAME

NOTES (OPTIONAL)

SCANNER-05

e.g. asset tag, freezer unit

Add scanner
Print setup cards

SCANNER	LINK			SIGNED IN	LAST SEEN	TEAM	NOTES		
SCANNER-01	http://127.0.0.1:3399/?d=jrwpm63	Copy	QR	9/25/2026	9/25/2026, 4:41:40 PM	1	freezer unit A	Reset link	Remove
SCANNER-02	http://127.0.0.1:3399/?d=4mutajzr	Copy	QR	9/25/2026	9/25/2026, 4:41:41 PM	2	freezer unit B	Reset link	Remove
SCANNER-03	http://127.0.0.1:3399/?d=3rjgn923	Copy	QR	9/25/2026	9/25/2026, 4:41:41 PM	3	freezer unit C	Reset link	Remove
SCANNER-04	http://127.0.0.1:3399/?d=chg52bzt	Copy	QR	9/25/2026	9/25/2026, 4:41:40 PM	—	spare, in the office	Reset link	Remove

On the device: open the link in Chrome (or scan the QR into the address bar), then menu → **Add to Home screen**. Opening the link signs that scanner in — the server accepts counts from signed-in scanners only. Treat a link like a key: **Reset link** issues a new one and kills the old, and removing a scanner stops it entirely. A link used more than once is flagged, which is normal after a scanner is wiped and worth a look otherwise.

Settings → Scanners. One row per handheld: its link, when it was last seen, Reset link and Remove.

On each handheld, once:

1. Open Chrome, scan the QR code from its card into the address bar (or type the link).
2. **Install it. Do not leave it running as a browser tab.** Chrome menu → **Install app / Add to Home screen**, or tap **Install on this scanner** on the app's own sign-on screen. Installed, it runs in its own window — no address bar at all — and starts even with no signal.

A browser tab is the one setup that causes trouble: Chrome keeps its address bar above the page, brings it back whenever the page moves, and if the keyboard focus ever leaves the page a scan is typed into that bar instead of into the count. The app warns you when it is running in a tab and offers **Hide the browser bar for this shift** as a stopgap.

Still want a locked-down device? See **Kiosk mode** at the end of this step.

1. Set up **DataWedge** so scans arrive as keystrokes: either as the end of a scan, and neither moves the cursor off the box)*
 - the profile associated with Chrome → **Keystroke output: enabled**
 - **Basic data formatting** → send a suffix: **ENTER** *(or TAB — the app takes
 - **Barcode input** → enable the symbologies your labels use.

Kiosk mode. Three levels, strongest last:

	WHAT IT TAKES	WHAT THE COUNTER SEES
Installed app (do this)	Chrome menu → Install app, once per device	Its own window. No address bar, no tabs, no back button
Full screen for a shift	The button on the sign-on screen, one tap	The bar goes away until the app is closed. The browser says so once, with a banner carrying the address

	WHAT IT TAKES	WHAT THE COUNTER SEES
Locked device	Zebra Enterprise Home Screen (built into the device) or your MDM, set to launch this app and nothing else	The handheld does nothing but count. Survives a reboot

The first two are inside the app. The third is a device setting — it is configured on the handheld or through your MDM, not here, and it is the only one that stops somebody leaving the app altogether.

Treat a scanner link like a key. Reset link issues a new one and kills the old — use it if a device is lost or a link gets out. Removing a scanner stops it entirely. A link used on more than one device is flagged in the table; that is normal after a scanner is wiped and worth asking about otherwise.



Print setup cards: one per scanner, each with its own QR code and the three steps to do on the device.

Step 8a — A barcode test book, to practise on

Settings → Scanners → Barcode test book. A page of real **Code 128** labels to print, cut up and scan — the same symbology the racking uses, so a gun that reads the book reads the rack. Use it to train a crew at a desk, to check a new scanner reads properly, or to dry-run a count before the real one.

Two ways to fill it:

- **From this count** — pick **Bins** or **Pallets**, optionally one aisle, and how many. The codes come straight from the uploaded lists, so what you scan off the paper is what the gun expects.
- **From a spreadsheet** — ↓ **Excel template** downloads a workbook with four columns:

COLUMN	WHAT GOES IN IT
Type	BIN or PALLET — printed above the bars so a trainee knows which question it answers
Code	What the barcode carries, and what the gun will receive
Label	Free text under the code — the item, the zone, anything useful
Note	A second, smaller line

Fill it in, upload it back, and the book opens ready to print. The codes need not exist in any count — that is the point of practising.

Print at 100%. “Fit to page” shrinks the bars and a scanner will refuse them. Plain white paper only: glossy or coloured stock scatters the beam.

Barcode test book

real Code 128 labels to practise on

A page of bin and pallet barcodes to print, cut up and scan — for training a crew, or dry-running a count before the real one. The same symbology the racking uses, so a gun that reads the book reads the rack. Take the codes from this count, or fill in the spreadsheet with whatever you want to practise on.

FROM THIS COUNT

LABELS FOR

Bins

▼

aisle

F01 — or blank

▼

HOW MANY

40

▼

PER ROW

2

▼

Open the book

FROM A SPREADSHEET

Download the template, type or paste your codes into it, then upload it back. Columns: **Type** (BIN or PALLET), **Code** — what the barcode carries — **Label** and **Note**.

↓ Excel template

Choose File

No file chosen

Turn it into barcodes

The test book card: from the count itself, or from a spreadsheet of whatever codes you want to practise on.

Step 9 — Set up the counting screen

Settings → Scanner screen.

This is what a counter actually sees. The preview on the right is drawn at the real pixel size of the device (MC9090 is 240 × 320; MC9200 is 480 × 640) using the scanner's own stylesheet, so it cannot drift from reality.

- **The questions, in order.** Drag to reorder. The default asks for the pallet, then the quantity, then the bin. Counting **location-first** — bin, then what is in it — suits a team working a bay at a time. Lot code and expiry sit in the order too, and only appear on counts that ask for them (Part 3). A question can never be dragged off the list entirely.
- **Show contents after a pallet scan** — shows the SKU and description the ERP has for that pallet, so the counter can see they are at the right one.

- **Show the next bin in the aisle** — once a team starts a section, the gun tells them which bin comes next (001 → 002 → 003), so nothing is skipped. A bin holding several pallets keeps the guide until every tag in it is accounted for — see step 15.
- **The on-screen keyboard never appears by itself**, on any screen — not for the quantity, not for a clock-in number. The scanner is the keyboard. The **Keyboard** button on the counting screen, and **Type it instead** on the sign-on screen, bring one up for whoever has to type.
- **Update itself after a deploy** — on unless you turn it off. A scanner left running keeps the version it started with: these are installed apps that sit on a cradle overnight and are the same page in the morning. With this on, the gun compares what it is running against what the server is serving and reloads itself when it has fallen behind — **never mid-pallet, and never with counts still queued on the device**. If it is in the middle of a line it shows a green **Update ready** bar and waits for the line to finish; tapping the bar takes it at once. It reloads once for any one version: a gun that comes back still behind says *"update did not take"* rather than reloading in a loop halfway down an aisle.
- **Keep it upright** — on unless you turn it off. These are portrait handhelds held one-handed at a rack face, and a screen that flips halfway down an aisle is unusable. There is **one** way up: the installed app asks the device for it by name, and a gun flipped end over end is not "still portrait" as far as this app is concerned. Only an app that owns the screen (installed, or full screen) is allowed to ask at all — a browser tab is always refused — so when the handheld turns anyway, the app turns *itself* back: sideways gets a quarter turn, end over end gets a half turn, and either way the counting screen reads upright in the hand. It is only a picture being turned — scanning, tapping and the comments countdown all carry on exactly as before. Turn it off and the gun is left however the device turned it, with a yellow **Hold the scanner upright** strip across the top. Turning auto-rotate off on the handheld (an Android setting, not an app one) is still the tidiest fix of all.
- **Take the whole screen** — off unless you turn it on. It makes the app fill the display at sign-on, but the browser announces that with a bar carrying the site's address, right over the counting screen. **Installing the app** on the handheld (step 8) is the quiet way to lose the browser, and the one to prefer.
- **Keep the screen awake** — holds the display on while a team is counting, instead of it sleeping between bays.
- **Buzz on a good or bad scan** — useful with ear defenders.
- **Text size: Large** — for gloves and a freezer.
- **Re-key a quantity of at least** — anything this big has to be typed twice. 0 never asks twice.
- **Comments step moves on after N seconds** — how long the comments step waits before it goes to the next bin by itself. It ships at **2 seconds**: long enough to tap a reason, short enough that a counter with nothing to say is not standing there. Use **–** and **+** to change it, or type a number. **0** turns the clock off and waits for the counter. Typing or tapping a chip always stops the countdown, whatever it is set to.

Press **Save**. Scanners pick the change up within about half a minute, between pallets — nobody has to sign out.

Settings → Scanner screen. Drag the questions into the order your site counts in; the preview is the real MC9090 screen size.

Step 10 — Set the one-tap reasons

Settings → Scanners → What the scanners offer.

Counters wearing gloves in a freezer will not type. These are the buttons they tap instead, and the reasons a site needs are its own — "Blocked by a trailer" means something here and nothing anywhere else.

- **Comments** — offered on the last step, after the bin is scanned.
- **Override reasons** — offered when a pallet ID is not on the list and the counter is allowed to accept it anyway. **Other** is always offered on the gun as well, whatever you configure.

The time the comments step waits before moving on is set with the rest of the counting screen — see step 9.

Press **Save**; again, scanners pick it up within about half a minute.

What the scanners offer

the defaults

the one-tap reasons a counter can pick, without typing

The reasons a site needs are the site's own — "Blocked by a trailer" means something here and nothing anywhere else. A scanner picks up a change within about half a minute, without anybody signing out of it.

Comments

Offered on the last step, after the bin is scanned.

Damaged

Partial pallet

Mixed pallet

Label unreadable

Needs recount

Blocked / could not reach

Add

Override reasons

Offered when a pallet ID is not on the list and the counter is allowed to accept it anyway.

Label unreadable

New receipt, not on the report

Relabelled

Hand-written ID

Supervisor said to count it

Add

How long the comments step waits before moving on by itself is set on the [Scanner screen](#) tab, with the rest of what the counting screen does.

Save

Restore the defaults

Settings → Scanners → What the scanners offer. These are the buttons a counter taps instead of typing.

Part 3 — Setting up a count

Do this the day before, not on the morning of the count.

Step 11 — Create the count

Dashboard → Progress → Count session → Start a new count.

1. **Name** it something you will recognise later: `Q3 2026 wall-to-wall`.
2. **Type**:
 - **Full count** — teams work whole aisles. This is a wall-to-wall.
 - **Cycle count** — a batch of bins per day or week; see Part 6.
3. **Bin list** — attach it here, or upload it later in Settings.
4. **Inventory report** — what the ERP thinks is on hand. **Attach it**. With it you get variances, second counts and an ERP adjustment file. Without it the count still records what is there, but nothing is compared against anything.
5. Press **Create count**.

The new count inherits the rack drawing the last one used, so the map works immediately.

The screenshot displays the 'Inventory Count' dashboard for 'Front Royal'. The left sidebar contains navigation links: 'Dashboard', 'Cycle counts', 'Teams & crew', and 'Settings'. The main content area is titled 'Dashboard' and shows a 'Progress' section with a 'FULL: Q3 2026 wall-to-wall 13%' status. Below this, there are statistics for 'COUNT LINES' (2,356) and 'SCANNERS' (3). A table lists the active aisles and bins being scanned. The 'Count session' section at the bottom allows for configuration of the count, including settings for 'WHAT THE GUN ASKS', 'HOW THE FLOOR IS RUN', 'SECOND COUNTS', and 'ADJUSTMENTS & ACCURACY'.

TEAM	SCANNER(S)	EMPLOYEE(S)	ACTIVE AISLE	LINES	BINS	LAST SCAN
1	SCANNER-01	E1043, E1088	Freezer - Aisle F02	1151	892	4:38:30 PM
2	SCANNER-02	E1102, E1157	Freezer - Aisle F04	1027	797	4:32:30 PM
3	SCANNER-03	E1163, E1190	Freezer - Aisle F05	178	120	4:40:30 PM

The count is picked in the header, on every supervisor page. Each row carries its own progress, so two counts are never confused.

Step 12 — Check the guided setup

Settings → Getting started.

A checklist worked out from what is actually in the database, not from a box somebody ticked. It tells you what is still missing and takes you straight to the screen that fixes it. For a full count it wants: the bin list, the inventory report, registered scanners, the scanner screen, racking blocks, a crew roster and a team plan.

Getting a count started October spot check - full count

Worked out from what is actually in the database, not from a box somebody ticked. Pick the count at the top of the page; the steps change for a cycle count.

2 thing(s) still needed before anyone can scan.
The rest is recommended rather than required.

- Upload the bin list** needed to start Upload it
Every location in the warehouse. It is what a scanned bin is checked against, and it defines the aisles teams get assigned to.
nothing uploaded yet
- Upload the inventory report** recommended Upload it
What the system thinks is on hand. Without it a count still records what is there, but nothing is compared, so there are no variances and no second counts.
nothing uploaded yet
- Register the scanners** Change it
Each handheld gets its own link. The server only takes counts from a scanner that has signed in with one.
4 registered · 4 signed in
- Pick the rack drawing** Change it
Draws progress over the real floor plan instead of a schematic.
using front-royal
- Pair the aisles that share racking** recommended Pair them
Two teams either side of the same rack get in each other's way. Pairing them keeps only one team in a block at a time.
every aisle is its own block
- Add the crew and the teams** Change it
Lets the app check a team has the equipment for the levels it is given, and check who signed on.
8 on the crew list · 3 team(s)
- Queue the aisles onto teams** needed to start Queue them
A guided count sends each team to its next aisle. Without a plan the scanners have nothing to hand out.
nothing queued yet

A new count: two things stop anybody scanning, the rest are recommended. Every step links to the screen that fixes it.

Getting a count started Q3 2026 wall-to-wall - full count

Worked out from what is actually in the database, not from a box somebody ticked. Pick the count at the top of the page; the steps change for a cycle count.

This count is running — 1,809 bins have a count already.
Anything still open below can be filled in as you go.

- Upload the bin list** Change it
Every location in the warehouse. It is what a scanned bin is checked against, and it defines the aisles teams get assigned to.
13,673 bins in 28 aisles
- Upload the inventory report** Change it
What the system thinks is on hand. Without it a count still records what is there, but nothing is compared, so there are no variances and no second counts.
2,658 pallets · 1,809 bins carry a last-counted date
- Register the scanners** Change it
Each handheld gets its own link. The server only takes counts from a scanner that has signed in with one.
4 registered · 4 signed in
- Pick the rack drawing** Change it
Draws progress over the real floor plan instead of a schematic.
using front-royal
- Pair the aisles that share racking** Change it
Two teams either side of the same rack get in each other's way. Pairing them keeps only one team in a block at a time.
14 block(s) of more than one aisle
- Add the crew and the teams** Change it
Lets the app check a team has the equipment for the levels it is given, and check who signed on.
8 on the crew list · 3 team(s)
- Queue the aisles onto teams** Change it
A guided count sends each team to its next aisle. Without a plan the scanners have nothing to hand out.
8 aisle assignment(s)

The same checklist once that count is ready to run: every step green, and the count already running.

Step 13 — Set the count's options

Dashboard → Progress → Count session. These apply to the count picked in the header, and a scanner that is already counting picks up a change within about half a minute. The questions it asks change between pallets, never mid-line.

OPTION	WHAT IT DOES	SUGGESTED
Pallet ID check	<i>Validate — allow override with reason:</i> an unknown pallet can be accepted with a reason. <i>Validate — no overrides:</i> it cannot. <i>Accept any ID:</i> no checking, though duplicates are still blocked.	Validate with override
Guided by aisle plan	Teams are sent to their assigned aisle and warned when they scan a bin outside it. Off means anyone can count anything.	On for a wall-to-wall
Ask for comments	Adds the optional comments step at the end of each pallet.	On
Auto second counts	Raises a "go back and look again" task automatically when a line disagrees with the report.	On
Scanners start here	Every scanner lands on this count at sign-on. They can still pick another.	On, on count day
Ask for the lot code	Adds a LOT CODE question, checked against the report. Wrong lot is called out at the pallet.	On only if you track lots
Ask for the expiry date	Adds an EXPIRY question and flags anything already out of date.	On for frozen food
Recount over N units	A difference smaller than this raises no second count.	2–5 units
or over N %	...or at least this share of the expected quantity. Either threshold is enough.	5–10 %
Cap open	Stop raising automatic second counts once this many are open, so the list stays walkable.	50–100
Adjustments need approval	Every difference from the report has to be approved, with a reason code, before it reaches the ERP file. Off by default.	On for a wall-to-wall that is audited
Approve over N units	A difference smaller than this goes through without a signature.	5–10 units
or over N %	...or at least this share of what the report expected. Either threshold is enough.	5–10 %

OPTION	WHAT IT DOES	SUGGESTED
ABC classes & accuracy	Reports accuracy by ABC class against a target for each. Off by default.	On when somebody reports on accuracy
Map drawing	Which rack drawing the map uses. <i>Schematic</i> builds one from the bin codes.	Your drawing

Why the thresholds matter. With both at 0 the app sends somebody back for a one-unit difference, and by mid-morning the second-count list is longer than the count. Set them once and the list stays short enough that people actually walk it.

Count session

#1 · full count · open

Settings for the count picked in the header. A scanner already counting picks up a change within about half a minute; the questions it asks change between pallets, never mid-line.

WHAT THE GUN ASKS

PALLET ID CHECK

Validate — allow override with rei ▾

☒ Ask for comments
 ☒ Ask for the lot code
 ☒ Ask for the expiry date

HOW THE FLOOR IS RUN

☒ Guided by aisle plan
 ☒ Scanners start here

MAP DRAWING

Front Royal - rack layout D-22P91 ▾

SECOND COUNTS

☒ Raise them automatically

OVER

3 units

OR OVER

5 %

CAP OPEN

60

Both at 0 sends somebody back for a one-unit difference.

ADJUSTMENTS & ACCURACY

☒ Adjustments need approval
 ☒ ABC classes & accuracy

APPROVE OVER

5 units

OR OVER

5 %

Save

Close session

Delete...

Start a new count

NAME

e.g. Q3 2026 wall-to-wall

TYPE

Full count — teams work whole aisles ▾

BIN LIST — optional, you can upload it later

INVENTORY REPORT — what the count is validated against

Choose File No file chosen

Choose File No file chosen

Create count

The inventory report is what the system thinks is on hand. With it you get variances and second counts; without it a count still records what is there, but nothing is compared. Either file can wait — [Getting started](#) tracks what is still missing.

Dashboard → Progress → Count session: every option above, on one card, for the count picked in the header.

Step 14 — Give the teams their aisles

Dashboard → Team plan.

Either upload a **counting plan** (Settings → Lists & racking → Counting plan: one row per aisle, in the order each team counts it, with the levels their equipment reaches), or queue aisles by hand here.

- A team's next aisle starts automatically as soon as the block it belongs to is clear. That is what staggers the crews apart.
- A team is not sent where its equipment cannot reach: if the roster says a crew is on foot, they will not be queued onto level D. A supervisor can override this.

- Two teams **can** share one aisle as long as their levels do not overlap — a forklift crew high and a crew on foot low.

Team assignments

Queue aisles per team in counting order, with the **levels** that team's equipment covers (A-C on foot, D-F with the lift, A-F for all). Levels are required. Two teams can share an aisle on different levels. A team's next aisle waits while another team holds those levels in its racking block — that is what staggers them. Which aisles share racking is set under [Settings](#).

TEAM	AISLES, IN ORDER	LEVELS (REQUIRED)
3	F01, F03, F05	A-C

Queue

Team 1 in Freezer - Aisle F02, levels A-F

Freezer - Aisle F01 - levels A-F **reopen** Freezer - Aisle F02 - levels A-F **done release** Freezer - Aisle F07 - levels A-F **start** **x**

Team 2 in Freezer - Aisle F04, levels A-F

Freezer - Aisle F03 - levels A-F **reopen** Freezer - Aisle F04 - levels A-F **done release** Freezer - Aisle F08 - levels A-F **start** **x**

Team 3 in Freezer - Aisle F05, levels A-F

Freezer - Aisle F05 - levels A-F **done release** Freezer - Aisle F06 - levels A-F **start** **x**

Dashboard → Team plan: each team's aisles in counting order — what they are on now, and what is queued behind it.

Inventory Count

Front Royal

Q Search anything...

- Dashboard
- Cycle counts
- Teams & crew**
- Settings

Teams & crew

8 people · 3 teams

Crew & teams Equipment rules

Crew list clock in number, name, department, equipment

CSV OR EXCEL FILE

Choose File No file chosen Replace the whole list **Upload** Sample CSV Export current list

Columns matched by name: **Clock in Number** (also accepts Badge, Employee ID), then Name, Department, Equipment. Equipment can be a list — Scissor lift, High reach, Spellings like forklift, reach truck, pallet jack are understood.

CLOCK IN NUMBER	NAME	DEPARTMENT	EQUIPMENT
E1042	Full name	Freezer	☒ Dock truck ☐ Scissor lift ☐ High reach truck

Add / update

drag people between the crew list and a team — or use the menu on a card

NEW TEAM	NOTES
3	e.g. night shift, freezer

Add team

Not on a team 2

Drag someone onto a team to assign them.

Nadia Haddad — no team — **Remove**

E1204 Warehouse - reaches level A

On foot

Owen Blackwell — no team — **Remove**

E1219 Warehouse - reaches level A

Dock truck

Team 1 reaches levels A-F

Between them: High reach truck, Scissor lift, Dock truck, On foot

Marcus Obi Team 1 **Remove**

E1043 Warehouse - reaches levels A, C, D, E, F

High reach truck Scissor lift

Priya Raman Team 1 **Remove**

E1088 Warehouse - reaches level A

Dock truck On foot

Delete team

Team 2 reaches levels A-F

Between them: Dock truck, On foot, High reach truck, Scissor lift

Ava Delgado Team 2 **Remove**

E1157 Warehouse - reaches level A

Dock truck On foot

Tom Zielinski Team 2 **Remove**

E1102 Warehouse - reaches levels A, C, D, E, F

High reach truck Scissor lift

Delete team

Teams & crew (/teams): the crew list and who is on which team. Equipment is what decides which levels a team can be sent to.

Part 4 — Counting day

Step 15 — The counter's procedure (on the handheld)

There is a film of this. <docs/gun-demo.webm> (about four minutes) is a recording of the real app on real data, working through everything below: signing on, counting a pallet, a bay with three tags, two labels on one pallet, a label that will not scan, an empty bin, a pallet nobody expected, a message from the office, voiding a line, losing the wifi, and handing the aisle back. Open it in any browser. It is worth showing a new crew before their first shift.

Signing on

1. Open the app from the home screen. It shows the scanner's own name.
2. Pick the **count session** (it lands on the default one).
3. Enter the **team number**.
4. Scan or type every **clock-in number** on the crew, pressing Enter after each. Tap a number to remove it.
5. Press **Sign on & load list**. The handheld downloads the bin and pallet lists for that count and can work from then on **with no signal**.

The screenshot shows the 'Inventory Count' app interface. At the top, it says 'SCANNER-01' and 'online'. Below that, it asks 'WHAT ARE YOU DOING?' with two buttons: 'Full count' (highlighted in blue) and 'Cycle count'. Then it asks 'WHICH ONE' with a dropdown menu showing '#1 — Q3 2026 wall-to-wall'. Below that, it asks 'TEAM NUMBER' with a text input field containing '1'. Then it asks 'CLOCK IN NUMBERS — SCAN THEM, THEN ENTER' with a 'Scan a badge' button and an 'Add' button. Below that, it says 'Type it instead — show the keyboard' with two buttons: 'E1043 ×' and 'E1088 ×'. Finally, it says 'Everyone on the crew — tap a number to remove it.' with a 'Sign on & load list' button (highlighted in blue) and a 'Refresh sessions' button.

Signing on: the count, the team number, and every clock-in number on the crew.

Counting a pallet

The gun asks one question per screen, in the order the site configured:

1. **Scan PALLET ID** — it shows what the ERP says is on that pallet.
2. **Enter QUANTITY** — big numbers may have to be typed twice.
3. **Scan LOT CODE / Enter EXPIRY** — only on counts that ask for them. A lot that disagrees with the report, or a date already past, is called out on the spot.
4. **Scan BIN LOCATION** — it says where that bin is and which face it is on.
5. **Comments** — optional. Tap a reason or type a note; leave it and the gun moves on by itself after a couple of seconds (Settings → Scanner screen sets how long). Typing or tapping a reason stops the countdown.

The line is saved on the handheld the moment the last question is answered, and pushed to the server whenever there is a signal. The header shows **online** / **OFFLINE** and how many lines are still queued.

After sign-on — the aisle this team has been given, what is queued behind it, and any second counts waiting

Question 1 — Next bin tells the team which bin comes next in the aisle, so nothing gets skipped

Question 2 — the gun shows what the ERP says is on that pallet, so the counter can see they are at the right one

Questions 3 and 4 — only on counts that ask for them. A lot that disagrees with the report is called out here, not a week later in a report

The last required question — the gun says where that bin is and which face it is on

Comments — tap a reason rather than type one. Left alone, the gun counts down and moves to the next bin by itself

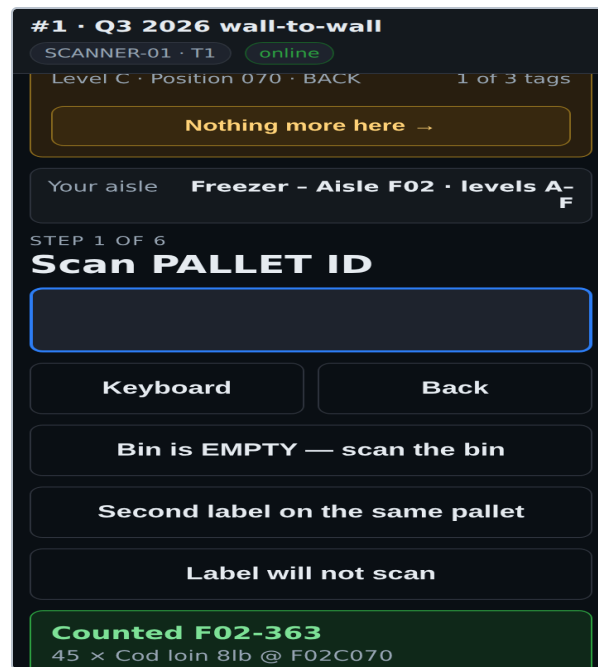
A bin with more than one pallet in it

Four pallet tags in one position is ordinary, and the guide stays put until they are all counted. After the first tag the banner turns amber and reads **Still in this bin** with a tally — *2 of 4 tags* — taken from the inventory report, so the team knows what is left to find before they walk on.

- Scan the next tag in the same bin. The tally goes up; when the report's pallets are all accounted for, the banner turns back to **Next bin** on its own.

- If the report is wrong and there is nothing more there, tap **Nothing more here** →. The bin is closed and the guide moves on.
- A bin the report lists nothing for stays open after a count, for the same reason — there may be a second pallet in it that the ERP has never heard of. Tap **Nothing more here** → when it is clear.
- **Bin is EMPTY** closes a bin outright.

The tally counts what the whole team has done, not just this scanner, so two handhelds working the same aisle never double-walk a bay or leave one half counted.



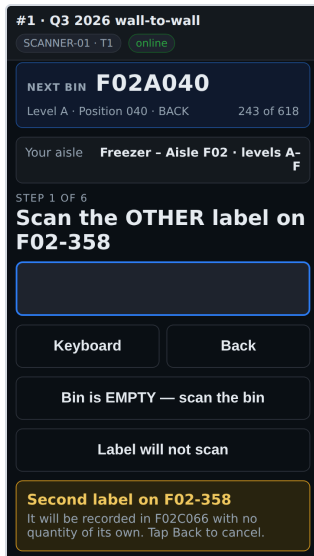
One tag into a position that holds three: the banner turns amber and stays on the bin until the other two are counted, or somebody says there is nothing more there.

A pallet with two labels on it

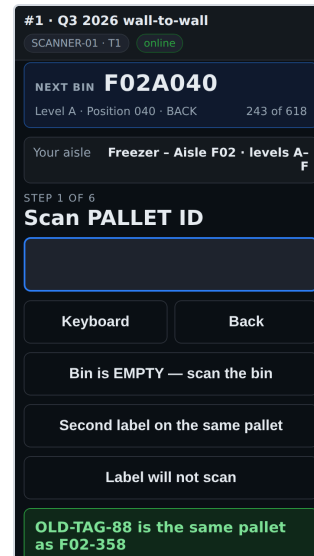
A re-tagged pallet that kept its old label, or one built from two, has two barcodes and is one pallet. Counting both as pallets doubles the stock; ignoring one leaves a live label for the next team to find.

1. Count the pallet once, as normal.
2. Tap **Second label on the same pallet**.
3. Scan the other label.

The tag is recorded against the pallet it is stuck to, with no quantity of its own. It shows on the pallet report as **SECOND LABEL**, the pallet it belongs to says *also tagged*, nobody is sent back for it, and it never reaches the ERP file as an adjustment. If the second tag is not on the inventory report, the gun offers the same answer in the "Pallet not on the list" screen.



The gun asks for the other label and names the pallet it will belong to



Recorded as the same pallet, with no quantity — so the stock is not counted twice

The green "Update ready" bar

A new version of the app is on the server and this scanner has not got it yet. Nothing is wrong and nothing needs doing: it loads itself the moment the pallet in hand is finished and everything counted has reached the server. Tap the bar if you would rather take it now.

When a label will not scan

Freezer labels come off, ice over, get clipped by a forklift. The pallet is still there and still has to be counted, so the gun has a way out that keeps the count going and tells a supervisor where to send somebody with a label printer.

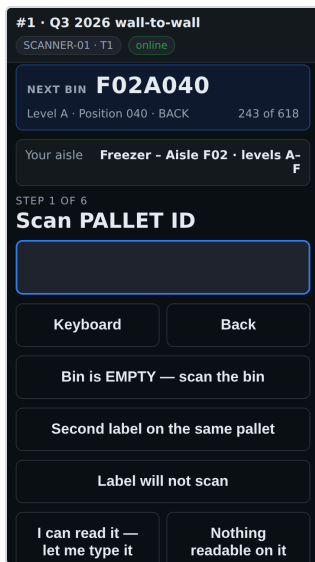
The same button is on both steps that read a label — **PALLET ID** and **BIN LOCATION** — and asks the same question: can you read it or not?

On the PALLET ID step — tap Label will not scan:

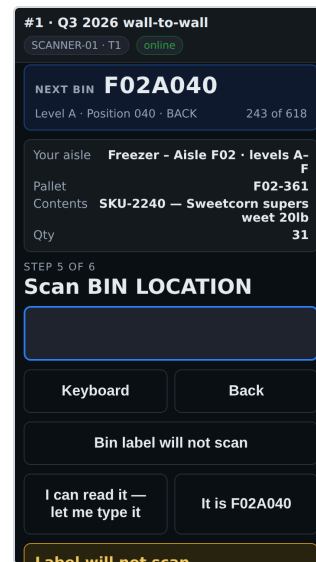
CHOICE	WHAT HAPPENS
I can read it — let me type it	The keyboard comes up, the counter types the number off the label, and counting carries on as normal. The line is flagged as a barcode to replace
Nothing readable on it	The pallet is counted anyway, under a name made from its bin — NO-LABEL - F01A001-1 . Give the quantity and scan the bin as usual

On the BIN LOCATION step — tap Bin label will not scan:

CHOICE	WHAT HAPPENS
I can read it — let me type it	The keyboard comes up and the counter types the bin code. It is checked against the bin list, the team's aisle and its levels exactly like a scan
It is F01A005	The app names the bin it believes the counter is standing at — the one the guide is on, the one they are already counting out of, or the one the report puts this pallet in — and one tap takes it. Checked the same way
No readable bin label	Only when the app has nothing to offer. The line is recorded against the aisle rather than being lost, and flagged as an unknown bin for a supervisor to sort out



On the pallet step: can you read it, or not?



On the bin step the app offers the bin it believes you are at, so nobody types a code off a rack leg in a freezer

Either way the pallet still gets counted — walking away from it is the one thing that would put the count out.

The line is marked either way, and the bin turns up on **Dashboard → Reports → Labels to replace** with what was counted there. A pallet with nothing readable on it counts as a pallet not on the report, so it also shows on the pallet report as **NOT IN MASTER** — which is exactly what it is until somebody puts a label on it.

Print the rack labels first. A pallet's bad label costs one counter one minute. A bin's costs every counter who walks up to that bay for the rest of the count — and the put-away driver afterwards. The relabel list marks them **RACK** and puts them at the top.

A message from the office

A supervisor can put a line on this team's scanners from the dashboard — *"come to the dock when you finish this aisle", "leave F12, the forklift is in it"*. It arrives within about twenty seconds, buzzes,

and sits on the screen until somebody taps **Got it**. An urgent one is red. Tapping Got it tells the dashboard who read it.



A message waiting on the counting screen. Nothing else is blocked — the scan box is still live underneath it.

The other buttons

BUTTON	WHEN TO USE IT
Keyboard	Brings the on-screen keyboard up to type an entry by hand — a number pad on the quantity step. It never appears on its own: the scanner is the keyboard, and a keypad covers half the screen. Tap it again to put it away
Back	Wrong entry — steps back one question
Skip	Lot, expiry or comments the pallet does not have
Bin is EMPTY	The bin is genuinely empty. Scan the bin; it is recorded as counted and empty
My aisle	Back to the assignment screen
History	The last 50 lines from this scanner. Void removes a wrong line from the totals
Aisle complete — next aisle	Only when the aisle is finished. It frees the racking block and releases the team's next aisle

#1 · Q3 2026 wall-to-wall

SCANNER-01 · T1 online

Recent counts

Most recent 50 lines from this scanner. Void removes a line from the totals.

F02-361 @ F02A040 4:43:12 PM · synced	31
VOID this line	
OLD-TAG-88 @ F02C066 4:43:05 PM · synced · second label on F02-358	0
VOID this line	
F02-358 @ F02C066 4:43:00 PM · synced	50
VOID this line	
F02-365 @ F02C070 4:42:55 PM · synced	59
VOID this line	
F02-364 @ F02C070 4:42:52 PM · synced	52
VOID this line	
F02-363 @ F02C070 4:42:48 PM · synced	45

History: the last 50 lines this scanner counted. **Void** takes a wrong one out of the totals.

When the gun asks for a reason

An unknown pallet, an unknown bin, a pallet already counted, a bin outside the team's aisle or level — the gun stops and asks why before it will take the line. Pick a reason from the list (or **Other**), add a note if it helps, and press **Accept and continue**. The line is saved and flagged for a supervisor. **Cancel — rescan** if it was simply the wrong barcode.

#1 · Q3 2026 wall-to-wall

SCANNER-01 · T1 online

Pallet not on the list

NO-LABEL-4471 is not in the uploaded pallet file.

Scanned NO-LABEL-4471

REASON (REQUIRED TO CONTINUE)

Choose a reason...

NOTE (OPTIONAL)

Free text

Accept and continue

It is a second label on F02-358

Cancel — rescan

#1 · Q3 2026 wall-to-wall

SCANNER-01 · T1 online

Pallet not on the list

NO-LABEL-4471 is not in the uploaded pallet file.

Scanned NO-LABEL-4471

REASON (REQUIRED TO CONTINUE)

Hand-written ID

NOTE (OPTIONAL)

Tag written by receiving

Accept and continue

It is a second label on F02-358

Cancel — rescan

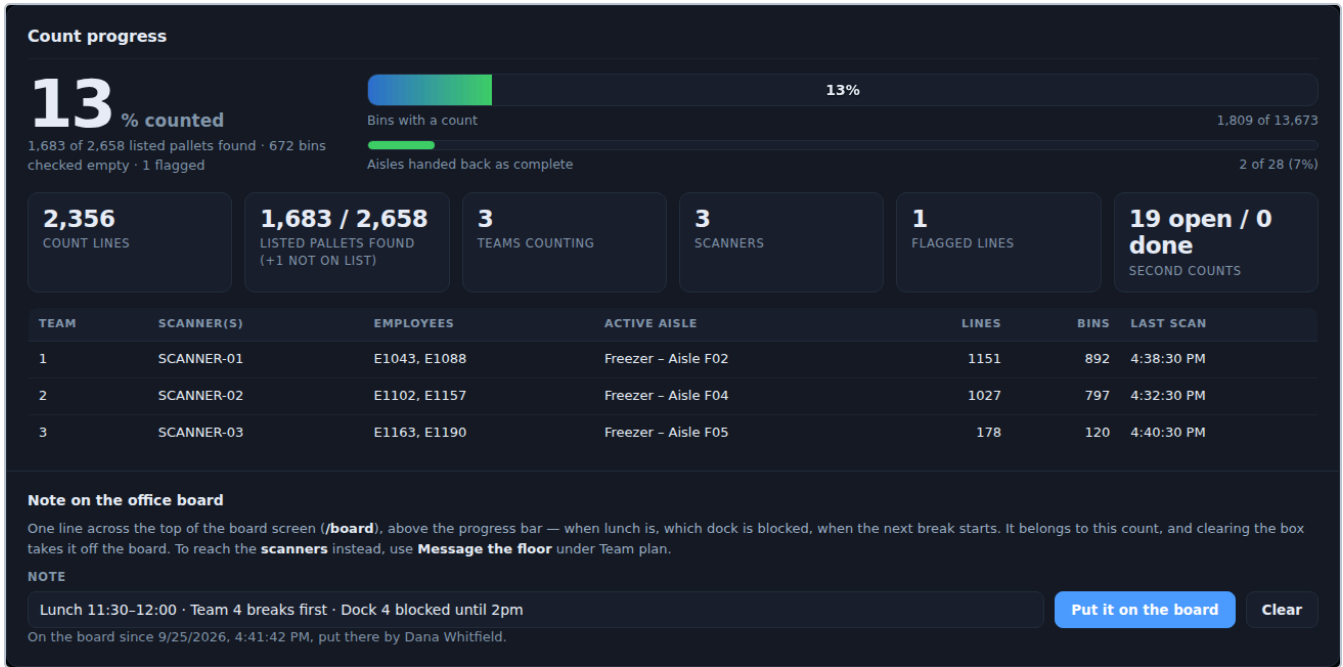
A pallet that is not on the inventory report: the gun will not take the line until somebody says why

The reasons in that list are the ones set in Settings, so they are your site's words. **Other** is always offered

Step 16 — The supervisor's procedure (on the dashboard)

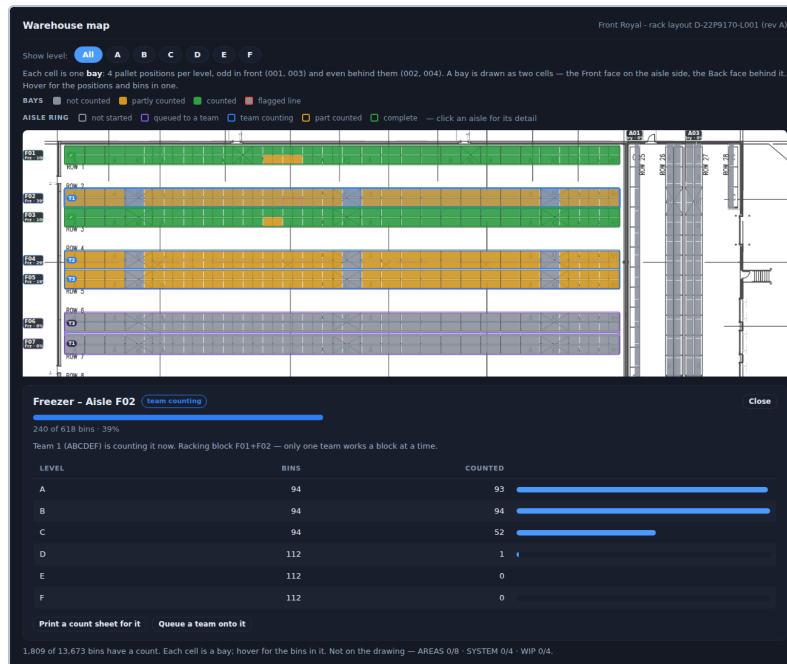
Dashboard, with the count picked in the header. It refreshes itself every 30 seconds.

- **Progress** — lines, bins counted, pallets, exceptions; and a row per team with what they are counting right now and when they last scanned. A team whose last scan was 40 minutes ago is a team with a problem.



Dashboard → Progress. The row per team is the one to watch: what they are on, how much they have done, and when they last scanned.

- **Map** — the warehouse from above. Aisles are outlined by state: not started, being counted now (with the team's number), done. Click an aisle for its bins, who counted it and what is flagged.



Dashboard → Map, with an aisle picked: 240 of 618 bins, level by level, and which team holds that racking block.

- **Team plan** — who is where, queue the next aisles, hand an aisle back. It also holds **Message the floor**: type a line, pick one team or all of them, tick **Urgent** if it cannot wait. The table underneath shows who has read each one — *2 of 3* means a scanner has not seen it yet, so do not assume. **Take it down** removes a message from the handhelds without losing what was said.

The 'Message the floor' interface allows sending messages to specific teams or all teams. It includes a 'TO' dropdown (set to 'Every team'), a 'MESSAGE' input field with the example 'e.g. Come to the dock when you finish this aisle', an 'Urgent' checkbox, and a 'Send' button. Below is a table of sent messages:

SENT	TO	MESSAGE	READ BY	
4:41:41 PM	every team	Fifteen minute break at 10:30, leave the guns on charge urgent	0 of 3	Take it down
4:41:41 PM	team 2	Skip F04 level D — the forklift is in it until 11	1 of 1 SCANNER-02	Take it down

Message the floor, under Team plan. "1 of 1" with the scanner named is a message that landed; "0 of 3" is one nobody has looked at yet.

- **Second counts** — see step 18.
- **Adjustments** — what this count would do to the ERP, and who signed for it. Only on counts with approvals turned on; see step 20.
- **Reports** — see step 19.

Watch for, during the count:

- Exceptions climbing on one team — usually a training problem, worth a radio call.

- A team stopped for a long time — dead battery, or stuck behind a trailer.
- An aisle nobody has started by mid-afternoon.

Step 16a — Finding anything: the search box

Every supervisor page, top of the sidebar. One box. Press / from anywhere on the page (or **Ctrl-K**) to jump into it.

It searches two things at once:

- **Your data** — a pallet ID, an item number, a word from a description, a bin code, a lot, an aisle, a name or clock-in number, a scanner, a count, a second count, an adjustment, something the office said to the floor, or a line in the log. Each hit carries the answer rather than just a link: a pallet shows what the report expected, what was counted, in which bin and by which team.
- **The app itself** — *"upload bin list", "keyboard", "approve", "racking blocks", "lunch note"*. These come back under **Go to** and take you to the card that does it, on whichever page it lives.

Choosing a hit takes you there: a tab on this page, or a page load that lands on the right tab. A lot code goes one better — it fills the lot box and runs it.

Inventory Count

Front Royal

Q HANDWRITTEN-F02X

COUNTED, NOT ON THE REPORT

HANDWRITTEN-F02

51 in F02A046 · team 1 · 2026-09-25 16:01

SECOND COUNTS

F02A046 · HANDWRITTEN-F02

NOT IN MASTER · open

ADJUSTMENTS

HANDWRITTEN-F02 +51

F02A046 · approved · Miscount — first count was wrong · sig...

Dashboard

FULL Q3 2026 wall-to-wall 13% ▾

ProgressMapTeam planSecond counts19Adjustmen

27-01-2926F06A004

27-01-3033F06A004

PALLET	SKU	DESCRIPTION	EXPECTED	1ST COUNT	COUNT
F01-008	SKU-6610	Salmon fillet skin-on 10lb	40		
F01-011	SKU-3350	Hash brown patty 240ct	61		
F01-017	SKU-4120	Chicken breast IQF 40lb	63		
F01-022	SKU-6610	Salmon fillet skin-on 10lb	58		

Typing a pallet ID: what the report expected and what was counted, the second count raised for it and the adjustment waiting to be signed — without opening any of those three cards.

Step 17 — The office board

Put `/board` on the office TV. It needs no sign-in and is read-only: the percentage counted, bins with a count, aisles handed back as complete, a row per team with what they are on, and every aisle as a

tile coloured by state. It deliberately shows no pallet IDs and no clock-in numbers.

It follows the newest open count on its own. To pin it to a particular one, add the session to the address: `/board?session=12`.

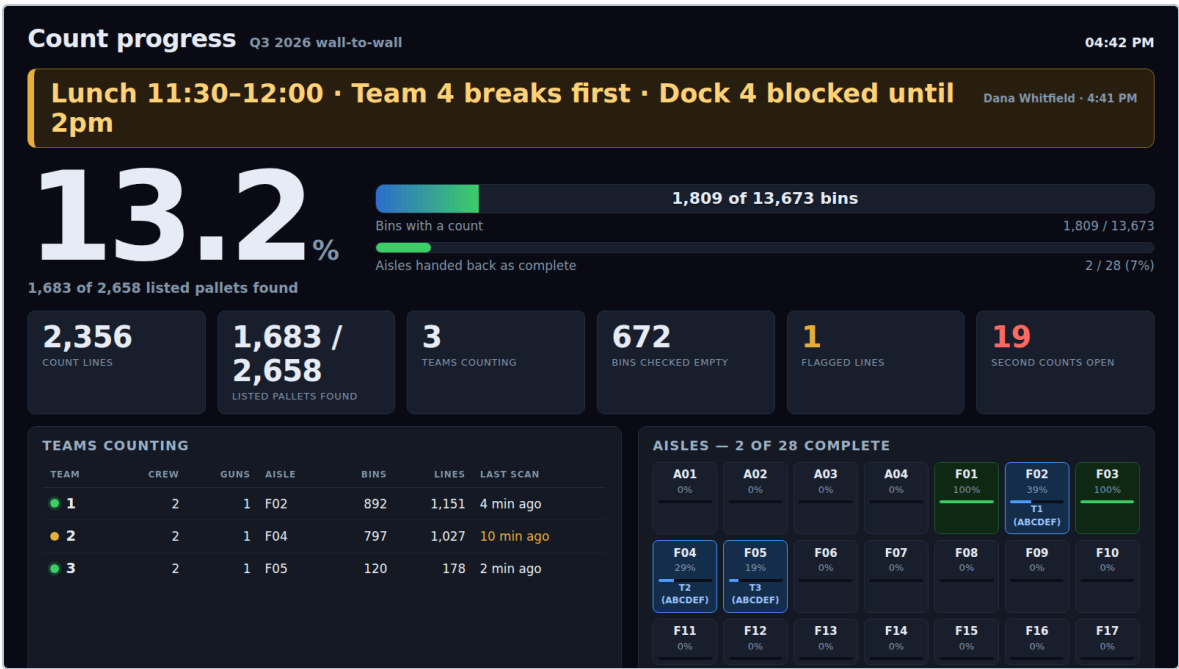
Lunch 11:30-12:00 · Team 4 breaks first · Dock 4 blocked until 2pm

Dana Whitfield · 4:41 PM

The note across the top of the board, above everything else — because it is the one thing somebody walking past is looking for.

The note across the top. Dashboard → Progress → Note on the office board writes one line that appears above the progress bar on the board, big enough to read from across the room: "Lunch 11:30-12:00 · Team 4 breaks first", "Dock 4 blocked until 2pm". It belongs to the count, it says who wrote it and when, and clearing the box takes it off the board.

Use the note for anything the floor reads walking past. To reach the **scanners** instead — one team, or all of them, with a read receipt — use **Message the floor** under Team plan (step 16).



The office board, for the TV: percentage counted, aisles complete, every team, and every aisle coloured by state.

Part 5 — Finishing a count

Step 18 — Work the second counts

Dashboard → Second counts.

A second count is a bin somebody has to walk back to. They are raised automatically when a line disagrees with the inventory report (subject to your thresholds), or by a supervisor.

- **Raise from all variances now** goes through everything that currently disagrees and raises the tasks. It deliberately leaves alone pallets in aisles nobody has counted yet — mid-count those are not missing, just not reached — and tells you how many it skipped and why.
- The tasks appear on the handhelds: a team taps **Start second counts** on the assignment screen, counts the bin again, and presses **Bin done — nothing more here**.
- A second count that agrees with the first settles the line.

Work these down before you close the count. A count closed with open second counts is a count with known-wrong numbers in it.

Second counts

19 open · 0 done

A second count is a bin somebody goes back to. They are raised automatically when a first count disagrees with the inventory report (quantity differs, wrong bin, not on the report, counted twice), when an aisle is handed back with expected pallets nobody saw, or by hand here. The counter is told the bin and the reason type, never the numbers, and the team that did the first count is never offered the second. Lines from a second count replace the first for that bin.

BIN

F03B012

or

PALLET ID

PLT03012B

NOTE

why

TEAM (OPTIONAL)

any

Request second count

Raise from all variances now

Export CSV

BIN	PALLET	REASON	DETAIL	SOURCE	1ST TEAM	TEAM	STATUS	1ST COUNT	2ND COUNT
F01A005	F01-008	QTY VARIANCE	expected 40, first count 26	auto	1	any	open	F01-008 ×26.0	done ×
F01A021	F01-024	QTY VARIANCE	expected 32, first count 41	auto	1	any	open	F01-022 ×58.0, F01-023 ×25.0, F01-024 ×41.0	done ×
F01C107	F01-053	WRONG BIN	expected in F01A041, found in F01C107	auto	1	any	open	F01-053 ×35.0, EMPTY ×0.0	done ×
F02A005	F02-008	QTY VARIANCE	expected 40, first count 26	auto	1	any	open	F02-008 ×26.0	done ×
F02A021	F02-024	QTY VARIANCE	expected 32, first count 41	auto	1	any	open	F02-022 ×58.0, F02-023 ×25.0, F02-024 ×41.0	done ×
F02D031	F02-052	WRONG BIN	expected in F02A040, found in F02D031	auto	1	any	open	F02-052 ×28.0	done ×

Dashboard → Second counts. The counter is told the bin and the reason type, never the numbers — and never the team that counted it first.

Step 19 — Read the reports

Dashboard → Reports.

- **Pallet report** — one row per pallet: expected vs counted, which bin the ERP expected and where it was actually found, and a status you can act on: `MATCH`, `QTY VARIANCE`, `WRONG BIN`, `MISSING`, `NOT IN MASTER`, `COUNTED TWICE`. Lot and expiry get their own columns, because the right count of the wrong lot is still wrong. **Only exceptions** narrows it to what needs attention — including a wrong lot or a date about to run out, even when the quantity is right.

Pallet report												
☒ Hide matching pallets				Pallet report CSV		Raw counts CSV		Exceptions CSV		Uncounted bins CSV		
PALLET	SKU	DESCRIPTION	EXPECTED	1ST COUNT	COUNTED	EXPECTED BIN	FOUND IN	LOT	EXPIRY	TEAM	COMMENTS	STATUS
F01-008	SKU-6610	Salmon fillet skin-on 10lb	40		26	F01A005	F01A005	L2026152	2027-01-31	1		QTY VARIANCE
F01-011	SKU-3350	Hash brown patty 240ct	61		61	F01A008	F01A008	L2026191	2026-10-07 soon	1		MATCH
F01-017	SKU-4120	Chicken breast IQF 40lb	63		63	F01A011	F01A011	L2026230	2026-09-05 expired	1		MATCH
F01-022	SKU-6610	Salmon fillet skin-on 10lb	58		58	F01A021	F01A021	L2026282	2026-10-07 soon	1		MATCH
F01-024	SKU-3310	Fries shoestring 6x5lb	32		41	F01A021	F01A021	L2026282	2027-02-16	1		QTY VARIANCE

Showing 500 of 1233 — export the CSV for the full list.

The pallet report with **Hide matching pallets** ticked: a short pallet, an expired date, one expiring soon. This is the list to work down.

- **Find a lot** — after a recall notice: type part of a lot code and get every case of it, both where it was actually counted and where the report expected it, so a pallet nobody found still shows up.

Find a lot

after a recall notice: where is every case of it

Searches what was actually counted, and what the inventory report expected — so it answers from the floor, not from what the ERP believed before the count started. Needs the lot code question turned on, or a report that carries lot codes.

LOT OR BATCH CODE

L2026139

Find it

15 counted, 18 on the report

Counted rows are where it actually is now.

PALLET	LOT	EXPIRY	QTY	BIN	AISLE	ITEM	TEAM	STATE
F01-005	L2026139	2027-01-28	59	F01A004	F01	SKU-2240 — Sweetcorn supersweet 20lb	1	done counted
F01-006	L2026139	2027-01-29	26	F01A004	F01	SKU-6610 — Salmon fillet skin-on 10lb	1	done counted
F01-007	L2026139	2027-01-30	33	F01A004	F01	SKU-6640 — Cod loin 8lb	1	done counted
F02-005	L2026139	2027-01-28	59	F02A004	F02	SKU-2240 — Sweetcorn supersweet 20lb	1	done counted
F02-006	L2026139	2027-01-29	26	F02A004	F02	SKU-6610 — Salmon fillet skin-on 10lb	1	done counted
F02-007	L2026139	2027-01-30	33	F02A004	F02	SKU-6640 — Cod loin 8lb	1	done counted

A recall: five cases of the lot found on the floor, and a sixth the report still expects in an aisle nobody has reached.

- **Count accuracy** — on counts with **ABC classes & accuracy** turned on: one row per class, and the whole count underneath. **Pallet accuracy** is how many pallets were exactly what the report said — the number an auditor asks for. **Quantity accuracy** is how many of the expected units were not in dispute. Each class is marked **MEETS** or **UNDER** against its target (Settings → Scanners → Accuracy targets; 99 / 97 / 95 % as shipped). Classes come from an **ABC** column on the inventory report; a report that has none can have them worked out from it with one button — Pareto on the expected quantities, the first 80 % of the stock A, the next 15 % B, the tail C. Anything the report classified itself is left alone.

Why by class. A warehouse can be 98 % accurate overall and still be losing money, because the misses are all on the fast movers. One number for the building hides that; a line for class A does not.

Count accuracy by ABC class, against the target for each

Pallet accuracy is how many pallets were exactly what the report said, and is the number an auditor asks for. **Quantity accuracy** is how many of the expected units were not in dispute. Classes come from an ABC column on the pallet file; if it has none, work them out from the report.

[Work out classes from the report](#) [Accuracy CSV](#) 2,658 of 2,658 pallets have a class

CLASS	PALLETS	EXACTLY RIGHT	PALLET ACCURACY	TARGET		UNITS EXPECTED	UNITS IN DISPUTE	QUANTITY ACCURACY	MISSING	NOT ON REPORT
Class A	1,385	1,377	99.4%	99%	MEETS	60,934	45	99.9%	0	0
Class B	298	291	97.7%	97%	MEETS	12,215	70	99.4%	0	0
Unclassified	1	0	0%	95%	UNDER	0	0	0%	0	1
Whole count	1,684	1,668	99%	95%	MEETS	73,149	115	99.8%	0	1

Bins with nothing odd in them: 1,808 of 1,809 (99.9%)

Count accuracy by class, with the whole count on the bottom row. Each class is held to its own target and marked MEETS or UNDER — and the one pallet with no class is the tag that was not on the report at all.

- **Labels to replace** — every label that would not scan, on a pallet or on the racking, with the bin to walk to, what was counted there, and whether there was a readable number on it at all. This is the walk-round with a label printer after the count. The **RACK** ones are listed first: everybody walks up to those.

Labels to replace pallets whose label would not scan

A counter who cannot scan a label — on a pallet or on the racking — says so on the gun and carries on counting. This is the walk-round afterwards: which bin, what was counted there, and whether there was a readable number on it at all. A pallet with nothing readable is counted under a name made from its bin, so it can be found again. **RACK** labels are the ones worth printing first: every counter after this one walks up to that bin too.

[Labels CSV](#) nothing reported — every label scanned

LABEL	BIN	AISE	COUNTED AS	PROBLEM	ITEM	QTY	TEAM	WHEN
-------	-----	------	------------	---------	------	-----	------	------

No label problems reported on this count.

The relabel list: which bin, what was counted there, and whether there was a number on it at all.

- **Count sheets** — printable paper sheets by aisle and level, for a dead battery or an auditor.

- **Exports** — the full count (every line as scanned), exceptions only, uncounted bins, the pallet report, second counts, the accuracy scorecard, the relabel list, and the adjustments with their reasons and signatures.

Step 20 — Approve the adjustments

Dashboard → Adjustments. Only on counts with **Adjustments need approval** turned on (step 13). With it off this tab says so, and every difference goes into the ERP file as it always did.

A variance is not an adjustment until somebody owns it. This is where that happens.

1. The list is every pallet that differs from the report, biggest first: what the system says, what was counted, and what the ERP would move.
2. Tick the ones you have satisfied yourself about. **Select all shown** takes the screenful.
3. Pick a **reason** — your site's list, set in Settings → Scanners → Adjustment reasons — and add a note if it helps.
4. **Approve selected**, or **Reject selected** if the count is wrong and the system should keep its number.

What the feature is actually for:

- **Nothing unsigned reaches the ERP.** The Adjustments export holds back anything still waiting, and the preview says how many it left out.
- **Anything under the threshold goes through unsigned.** A two-case difference on a pallet of 600 is noise; a count where every line needs a signature gets rubber-stamped, not read.
- **Counted again means signed again.** If a second count changes a pallet after it was approved, it comes back to this list with the approval cleared — the number somebody signed for is no longer the number.
- **A variance a second count clears disappears.** There is nothing left to adjust.
- **It is all in the log:** who approved what, with which reason, and when. That is the answer to "who authorised writing off 400 cases of chicken".

A pallet nobody has reached yet is not on this list while the count is running. Once the count is **closed**, a pallet the report expected and nobody found becomes an adjustment like any other.

Adjustments what this count would do to the ERP

Every pallet that differs from the inventory report, biggest first. Approve one with a reason code and it goes into the ERP file; reject it and the system keeps its number. Nothing leaves in the **Adjustments** export until it has been signed for — and a line counted again after it was approved comes back here, because the number somebody signed for is no longer the number.

10
WAITING FOR APPROVAL

1
APPROVED

0
REJECTED — THE SYSTEM KEEPS ITS NUMBER

0
UNDER THE THRESHOLD

115
UNITS STILL UNSIGNED

REASON: Choose a reason... NOTE (OPTIONAL): Free text — kept with the approval

☒ Only the ones still waiting Approve selected Reject selected Adjustments CSV

Select all shown Select none

	PALLET	ITEM	BIN	FOUND	SYSTEM	COUNTED	ADJUSTMENT	STATUS	REASON	SIGNED BY
☐	F01-008	SKU-6610	F01A005	QTY VARIANCE	40	26	-14	PENDING	—	—
☐	F02-008	SKU-6610	F02A005	QTY VARIANCE	40	26	-14	PENDING	—	—
☐	F03-008	SKU-6610	F03A005	QTY VARIANCE	40	26	-14	PENDING	—	—
☐	F04-008	SKU-6610	F04A005	QTY VARIANCE	40	26	-14	PENDING	—	—
☐	F05-008	SKU-6610	F05A005	QTY VARIANCE	40	26	-14	PENDING	—	—
☐	F01-024	SKU-3310	F01A021	QTY VARIANCE	32	41	+9	PENDING	—	—
☐	F02-024	SKU-3310	F02A021	QTY VARIANCE	32	41	+9	PENDING	—	—
☐	F03-025	SKU-6640	F03A022	QTY VARIANCE	39	48	+9	PENDING	—	—

10 shown of 11 · anything under 5 units and 5% goes through unsigned

The adjustments waiting for a signature, biggest first. AUTO is under the threshold and goes through without anybody being asked; APPROVED carries the reason and the name.

Step 21 — Send it to the ERP

Settings → ERP & backups → Send to the ERP.

1. Choose the **layout** that matches your ERP. Column names and which rows are included are configuration — add a layout rather than editing the file by hand afterwards.
2. **Preview** and read the first rows. On a count with approvals on, the preview says how many adjustments it left out because nobody has signed for them yet.
3. **Download CSV** and import it into the ERP.

The **Adjustments** layout carries the reason code and the name of whoever approved it, so the file the ERP gets is the same story the log tells.

Send to the ERP column names and rows are configuration — add a layout to match your system

LAYOUT: Pallet lines — one row per pallet counted

Preview Download CSV

Settings → ERP & backups. Preview before you download; the layout decides the column names and which rows are included.

Step 22 — Close the count

Dashboard → Progress → Count session → Close session. A closed count is read-only: scanners can no longer post lines to it, and its reports stay available for ever.

Delete... is separate and deliberately harder: the count must be closed first, you have to type its name, and a backup is taken before anything is removed. Everything counted against it goes with it. Close counts; delete only the ones created by mistake.

Step 23 — Backups and the log

Settings → ERP & backups → Backups & log.

- **Back up now** before and after anything large — an ERP import, a bulk re-upload, a deletion. The app also backs itself up daily and keeps the last 14.
- **The log** records every supervisor action: who signed in, who uploaded what, who overrode what, who raised second counts, who searched for a lot. **Export the log** for an auditor.

Backups & log

Back up nowExport the log

1 kept (newest 9/25/2026, 4:41:40 PM), one a day, 14 retained

BACKUP	SIZE	TAKEN		WHEN	WHO	DID WHAT
inventory-2026-09-25-16-41-40-daily.db	216 KB	9/25/2026, 4:41:40 PM	Download	9/25/2026, 4:41:44 PM	shared password	signed in with the shared password
				9/25/2026, 4:41:43 PM	Dana Whitfield	generated a cycle batch
				9/25/2026, 4:41:43 PM	Dana Whitfield	uploaded pallets
				9/25/2026, 4:41:43 PM	Dana Whitfield	uploaded bins
				9/25/2026, 4:41:43 PM	Dana Whitfield	created session
				9/25/2026, 4:41:43 PM	Dana Whitfield	created session
				9/25/2026, 4:41:42 PM	Dana Whitfield	put a note on the board

Backups on the left, the log on the right. Both export.

Part 6 — The cycle-count programme

A cycle count is the same app with a different rhythm: instead of stopping the warehouse, a handful of bins are counted every day.

1. **Create one count session of type "Cycle count"** and keep using it. It runs for the year — do not create a new one each week, or you lose the history of when each bin was last counted.
2. Upload the bin list and the inventory report. If your ERP export has a *last physical inventory date* column it is read automatically, and the programme picks up where the ERP left off.
3. Go to `/cycle` → **Today's bins**. Choose:
 - **how many bins** in the batch,
 - **which bins**: *longest since it was counted, never counted, or a random sample*,
 - optionally a **zone, aisle or levels** to stay inside.
4. **Preview**, then **Generate**. The bins appear on the handhelds as tasks.

`/cycle` → *Today's bins: how many, which ones, and optionally the zone or aisles to stay inside.*

1. Counters pick **Cycle count** at sign-on and work the list. Each bin is counted as it stands — it is *the* count for that bin, not a second opinion.
2. **Coverage** shows how much of the warehouse has been counted in the period, and what has not been touched. Export it for the auditors.
3. A **schedule** generates the batch automatically each day or week so nobody has to remember.

Inventory Count

Front Royal

Q Search anything...

Dashboard

Cycle counts

Teams & crew

Settings

shared password

Log out

Cycle counts

America/New_York · 2026-09-25

Today

Still open

Coverage

Program & data

Program

a cycle-count session runs for the year — keep using the same one

CYCLE COUNT

#3 — Cycle count programme 2026

Close

NEW PROGRAM

e.g. 2026 cycle counts

Create

Bin list & inventory report

The bin list is uploaded once. Re-upload the **inventory report** whenever you want current expected quantities — tick *Replace existing* so old pallets are cleared out.

FILE

Inventory report — pallet id, sku, description, qty.

CSV OR EXCEL

Choose File

No file chosen

☒ Replace existing

Upload

Load Front Royal bin list

The batches that have been generated, and the programme's own bin list and inventory report.

Physical Inventory Counting · SOP

40

Part 7 — When something goes wrong

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
Gun: <i>"this scanner is no longer authorised"</i>	Its link was reset or the scanner was removed	Settings → Scanners → Reset link , open the new link on the device
Gun: <i>"Offline and no list cached for this session"</i>	The handheld has never downloaded this count	Carry it into Wi-Fi once and sign on again
Gun says OFFLINE with lines queued	Normal in a dead spot	Nothing. They upload when it gets a signal. Do not wipe the device
A scan does nothing	DataWedge is not sending a suffix	DataWedge → Basic data formatting → send ENTER (or TAB)
A keypad covers the screen	Somebody left the Keyboard button on	Tap Keyboard again. It is off by default and never comes up by itself
The Adjustments tab is empty	Approvals are off for this count, or nothing differs from the report yet	Turn on Adjustments need approval under Count session. A pallet nobody has counted yet is not an adjustment until the count is closed
The ERP file is shorter than the variance list	Adjustments are waiting for approval and are held back deliberately	Work the Adjustments tab, then export again. The preview says how many were left out
Count accuracy says "no pallet has a class"	The inventory report has no ABC column	Add one (ABC, Class, Velocity — any of them is read), or press Work out classes from the report
A pallet is on the report as NOT IN MASTER with a name like NO-LABEL-F01A001-1	Its label would not scan and nothing on it was readable, so it was counted under the bin's name	Send somebody to relabel it — Reports → Labels to replace has the list
A line is in a bin called NO-LABEL-BIN-F01-1	The rack label would not scan, and the app had nothing to suggest — an unguided count, with the pallet not on the report either	The quantity is safe. Relabel the bay, then correct the bin on the line if it matters; the relabel list says which aisle it was in

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
The note is not on the board	The board polls every 15 seconds by default, or the note is on a different count	Wait a few seconds; check the board is pinned to the same count (<code>/board?session=12</code>)
A change you made does not appear on the scanners	The gun is still running the app it loaded before the change	Site settings (questions, reasons, text size) reach a running gun within about half a minute and need nothing. A new version of the app needs the page to reload: with Update itself after a deploy on it does that by itself between pallets, within a couple of minutes of somebody picking the gun up. To force it: tap the green Update ready bar, or close the app fully (not just the home button) and reopen it. Reinstalling is never necessary
The printed barcodes will not scan	The page was printed “fit to page”, or on glossy paper	Print again at 100% on plain white paper. If a gun still refuses them, print with Per row: 1 — the bars are wider on a bigger label
A code in the spreadsheet will not print	Code 128 carries plain ASCII only — an accented letter or a smart quote cannot be drawn	The book says which character it was; retype that code with plain letters and numbers
You cannot find where something is set	The app has four pages and twenty-odd cards	Type what you would call it into the search box at the top of the sidebar — <i>“upload bin list”</i> , <i>“keyboard”</i> , <i>“approve”</i> — and the Go to hits take you straight there. <code>/</code> puts the cursor in it
Not sure whether a gun has the latest version	—	The sign-on screen's bottom line reads <i>“App build a1b2c3d4e5f6 on the server — this scanner is up to date”</i> . Compare it across two guns, or against a fresh reload
A gun keeps saying “update did not take”	It reloaded and is still on the old version — something between it and the server is serving stale files	Close the app fully and reopen it. If it persists, clear the site data for the app on that device (Android → Settings → Apps → the app → Storage), then open its link again

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
The screen keeps rotating, or reads upside down	Auto-rotate is on, and a browser tab is never allowed to lock the orientation	The app turns itself back upright on its own, sideways or end over end. If it is not, check Keep it upright is ticked under Settings → Scanner screen, then read the grey line at the bottom of the sign-on screen — it says the screen size the handheld gave the app, how far the device says it has turned, and what the app did about it. To stop the device turning at all, install the app (it asks for one way up) or turn auto-rotate off on the handheld
A team says they never got a message	Look at Read by on the dashboard	It shows which scanners have tapped Got it. A scanner that is offline gets it on its next sync
A scan seems to land on a button instead of the box	Something else took the focus	Nothing — the app puts the keystrokes in the box and carries on. Tell us if it still happens
The guide is a bin or two ahead of the team	The bins hold several pallets each	Fixed: the guide now stays on a bin until its tags are counted. Check Show the next bin in the aisle is on
A scan opens the address bar and the text goes into it	The page has lost the keyboard — either the scanner sends a TAB that used to move focus out of the page, or somebody tapped the browser's own bar	Fixed: TAB now ends a scan like ENTER and the focus never leaves the box. If it ever happens again the app shows a red Tap here to scan bar — one tap puts it right
A bar with the web address appears on every scan	The app is being run as a page in Chrome rather than installed	Install it: Chrome menu → Install app , or the Install on this scanner button on the sign-on screen, then open it from the home-screen icon
The browser's address bar is in the way	Same thing — the app is not installed	As above. For a device that should run nothing else, use Zebra's Enterprise Home Screen
Gun: <i>"Team N is counting aisle X"</i>	Another team holds that racking block	Wait, or hand the other aisle back first
Second-count list is enormous	Thresholds are at 0	Set Recount over and or over % , and a cap (Part 3, step 13)

WHAT YOU SEE	WHAT IT MEANS	WHAT TO DO
The map is a schematic, not your drawing	No rack drawing on this count	Dashboard → Progress → Count session → Map drawing
A reason code edit "did not reach" a gun	It takes up to about half a minute, and only lands between pallets	Wait for the counter to finish the pallet they are on
Can nobody sign in?	All logins lost	The shared password turns itself back on when there are no admin accounts; the superadmin login is recreated at start-up
Numbers look wrong after an ERP import	The inventory report moved on	Re-upload it with Replace what is there , then re-read the pallet report

Part 8 — Capacity: what the system will take

Measured against the real Front Royal bin list (13,673 bins), with fifteen teams and thirty scanners counting simultaneously while the office board and two dashboards refreshed over the top of them:

Scanners posting at once	30, no errors, nothing lost, nothing double-counted
Throughput	~1,200-1,600 count lines per second
A scan, under that full load	75 ms median, 205 ms at the 95th percentile
Office board / dashboard refresh	~140 ms median while all thirty guns were writing
Whole master list to a handheld	~0.4 s each, thirty pulling simultaneously

Thirty scanners is comfortably inside what the server will take — real counters scan a pallet every ten to twenty seconds, which is a small fraction of the load above. The constraints that matter in practice are warehouse Wi-Fi coverage and battery life, not the server.

This is checked automatically: `node tests/run-all.mjs load-15-teams` runs the whole scenario and fails if anything is lost, doubled, mis-attributed, or if two teams ever end up in one racking block.

Appendix A — File column reference

Files may be CSV, TSV or Excel (.xlsx/.xlsm). Column names are matched loosely — case, spaces and punctuation are ignored — so most ERP exports load unchanged. Each upload card shows exactly which columns it recognised.

Bin list — the only required column is the location.

WHAT	ACCEPTED COLUMN NAMES
Bin (required)	location, loc, bin, bin location, location code, slot, code, warehouse location
Aisle	aisle, row, aisle no, aisle number — <i>derived from the bin code if absent</i>
Zone	zone, area, section, region, warehouse
Level	level, levels, tier, shelf — <i>derived from the bin code if absent</i>
Description	description, desc, name
Last counted	last phys invt date, last counted, last count date, last inventory date

Inventory report — the only required column is the pallet ID.

WHAT	ACCEPTED COLUMN NAMES
Pallet (required)	pallet id, pallet, container, container id, LPN, licence plate, pallet no, id, tag
SKU	sku, item, item number, item code, part number, product, material, stock code
Description	description, desc, item description, product name
Quantity	qty, quantity, on hand, on hand qty, expected, expected qty, system qty, cases, units
Unit	uom, unit, unit of measure, um
Location	<i>same names as the bin list</i>
Lot	lot, lot code, lot no, lot number, batch, batch code, batch number
Expiry	expiry, expiry date, expiration, expires, best before, use by, shelf life date
ABC class	abc, abc class, abc code, class, item class, velocity, velocity code, movement class, category — A , B or C , upper or lower case

Barcode test book — one row per label to print.

WHAT	ACCEPTED COLUMN NAMES
Type	type, kind — <code>BIN</code> or <code>PALLET</code>
Code (required)	code, value, barcode, bin, pallet, pallet id, bin location
Label	label, description, desc
Note	note, notes

Counting plan — one row per aisle, in the order the team counts it.

WHAT	ACCEPTED COLUMN NAMES
Team	team, team number, crew, group
Aisle	aisle, row, aisle no
Levels	level, levels, tier, shelf — e.g. <code>A-C</code> , <code>D-F</code> , <code>A-F</code>

Dates are read in either order — `2027-03-15` and `03/15/2027` both work.

Appendix B — Server settings reference

SETTING	DEFAULT	WHAT IT DOES
ADMIN_PASSWORD	changeme	The shared supervisor password. Change it
SHARED_PASSWORD_LOGIN	on	off requires named logins — ignored if no admin account exists, so it cannot lock you out
SUPERADMIN_USER	—	Username of the permanent admin, created at start-up
SUPERADMIN_NAME	—	That person's name, as it appears in the log
SUPERADMIN_PASSWORD	falls back to ADMIN_PASSWORD	Must be 8+ characters and not changeme
SCANNER_AUTH	on	off lets anything on the network post counts. Leave it on
DB_PATH	data/inventory.db	Put this on a persistent volume
BACKUP_DIR	data/backups	Where the daily backups go
BACKUP_KEEP	14	How many backups to keep
SITE_TIMEZONE	America/New_York	Dates on reports and cycle-count due dates
PORT / HOST	3000 / 0.0.0.0	Where the server listens
MAX_UPLOAD_MB	64	Largest file an upload will accept